Statistics refresher and linear regression

International Marketing Analytics

Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at
Office hours: book online (Google Calendar)

SESSION 1 · WT 2026/27

Today

Refresher → Python by hand → OLS theory → Tests → Chocolate in Python → In class, exercise

Refresher, Parts 1 to 5

- Scale types: what you may calculate
- Mean, median, standard deviation, standard error
- z-scores and the normal distribution
- Covariance and correlation
- By hand in Python; the kiosk's year; quiz 1

Regression, Parts 6 to 11

- OLS, inference, fit, logs and elasticities
- Seven assumptions and four tests
- Regression in the real world
- statsmodels and plotnine on real data; quiz 2
- In-class assignment, coding exercise 1

Readings: Bojinov, Parzen & Hamilton (2025), Linear Regression (HBS); Skiera, Reiner & Albers (2022), Regression Analysis.

Our example: a chocolate kiosk

Five weeks, five prices

Week	Price (EUR)	Bars sold
1	2.00	110
2	2.50	100
3	3.00	80
4	3.50	70
5	4.00	40

One week, day by day

Day	Bars sold
Mon	10
Tue	12
Wed	11
Thu	13
Fri (event)	54



Small on purpose: every number can be checked with a calculator, then in Python. Later: the kiosk's full year, 52 weeks.

Scale types

Part 1 of 11 · nominal, ordinal, interval, ratio

Four scale types

Scale	What you can say	Kiosk example	Typical statistics
Nominal	same or different	brand, payment method	counts, mode
Ordinal	more or less	rating 1 to 5 stars, size S/M/L	median, percentiles
Interval	how much more (no true zero)	temperature in °C	mean, standard deviation
Ratio	how many times (true zero)	price, bars sold, revenue	all, incl. ratios

Two groups: nominal and ordinal are categorical; interval and ratio are metric (numbers you can average).

In regression: categorical variables enter as dummies (0/1), metric variables as they are.

Why the scale matters

20 °C is not twice as warm as 10 °C: no true zero

Mean of star ratings: treats ordinal as metric; say so

Store ID 17: a number, but only a label: nominal

Quick quiz: which scale?

- Country of the customer
- Satisfaction from 1 to 7
- Age in years
- Discount: yes or no
- Temperature on the day

Rule: check the scale before you average, chart or put a variable into a model.

Centre and spread

Part 2 of 11 · mean, median, variance, standard deviation, standard error

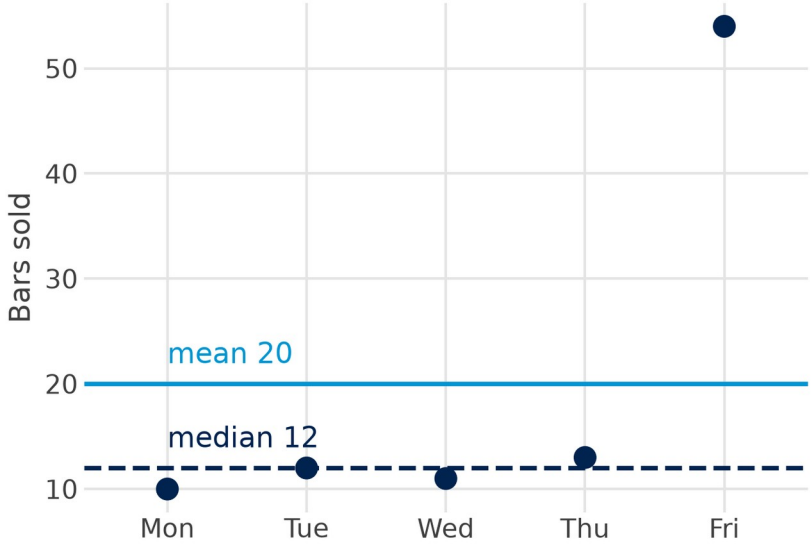
Mean and median

```
mean    x̄ = Σ x / n = 100 / 5 = 20
median  middle of 10 11 12 13 54 = 12
```

Mean uses every value; one event day pulls it up

Median is robust: half the days below, half above

Use the median for skewed data and ordinal scales



Totals need the mean: 20 bars a day × 5 days = 100 bars in the week.

Variance and standard deviation, by hand

Week	Bars x	$x - \bar{x}$	$(x - \bar{x})^2$
1	110	30	900
2	100	20	400
3	80	0	0
4	70	−10	100
5	40	−40	1,600
Sum	**400**	**0**	**3,000**

Variance and standard deviation

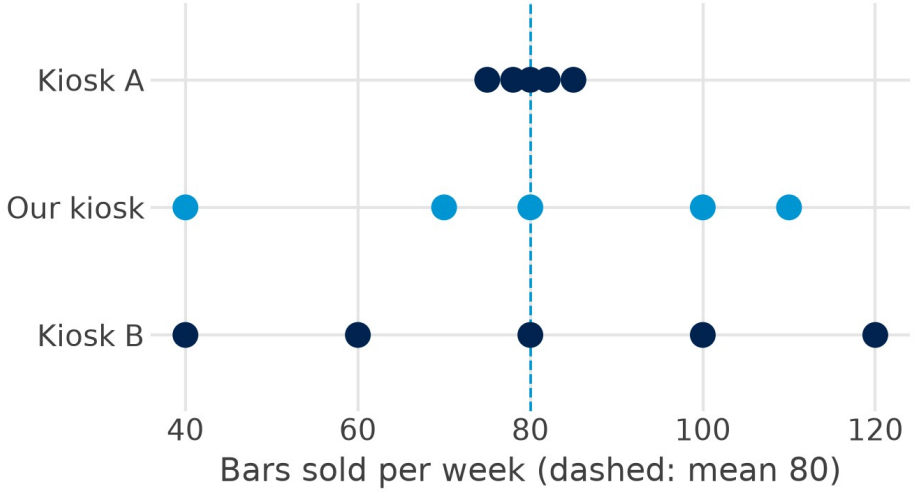
$\bar{x} = 400 / 5 = 80$
$s^2 = 3{,}000 / (5 - 1) = 750$
$s = \sqrt{750} = 27.4$ bars

Variance is in squared units (bars²)

Standard deviation is in bars: the typical distance from the mean

Σ **Why n − 1?** deviations from the sample mean are a little too small; dividing by n − 1 corrects this. polars and pandas do it by default.

Same mean, different spread



3.8 sd of kiosk A, weeks of 75, 78, 80, 82, 85 bars

27.4 sd of our kiosk, weeks of 110, 100, 80, 70, 40 bars

31.6 sd of kiosk B, weeks of 40, 60, 80, 100, 120 bars

For the manager: all three sell 80 bars a week, but kiosk B needs much more safety stock and its forecasts are less certain.

Standard error: how precise is the mean?

```
SE = s / √n          rough 95% interval ≈ x̄ ± 2 · SE
```

12.2

SE with our 5 weeks: 27.4 / √5, in bars

6.1

SE with 20 weeks: 27.4 / √20, in bars

55–105

rough 95 % interval for the mean: 80 ± 2 · 12.2 bars

sd = 27.4 describes the weeks: a single week is typically 27 bars away from 80

SE = 12.2 describes the average: it shrinks with √n; four times the weeks, half the SE



In every regression table: the column 'std err' is the standard error of a coefficient.

Standardisation and the normal distribution

Part 3 of 11 · z-scores, normal and standard normal distribution

Standardisation: z-scores

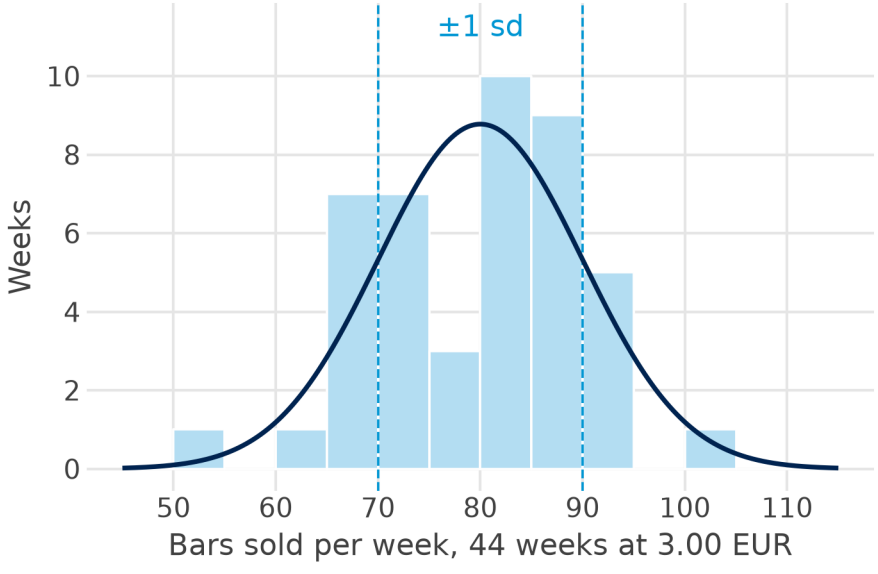
```
z = (x − x̄) / s        a standardised variable has mean 0 and standard deviation 1
```

Kiosk (spread slide)	Bars this week	Mean	sd	z
Kiosk A, steady	85	80	3.8	+1.3
Kiosk B, volatile	100	80	31.6	+0.6
Our kiosk, week 5	40	80	27.4	−1.46

Read it: 85 bars at steady kiosk A lie 1.3 standard deviations above its mean, 100 bars at kiosk B only 0.6: kiosk A's week was the more unusual one.

Use: compare across kiosks, countries or units; spot outliers (|z| > 3); standardised coefficients.

The normal distribution



68 % ±1 sd, 70 to 90 bars: 31 of our 44 weeks

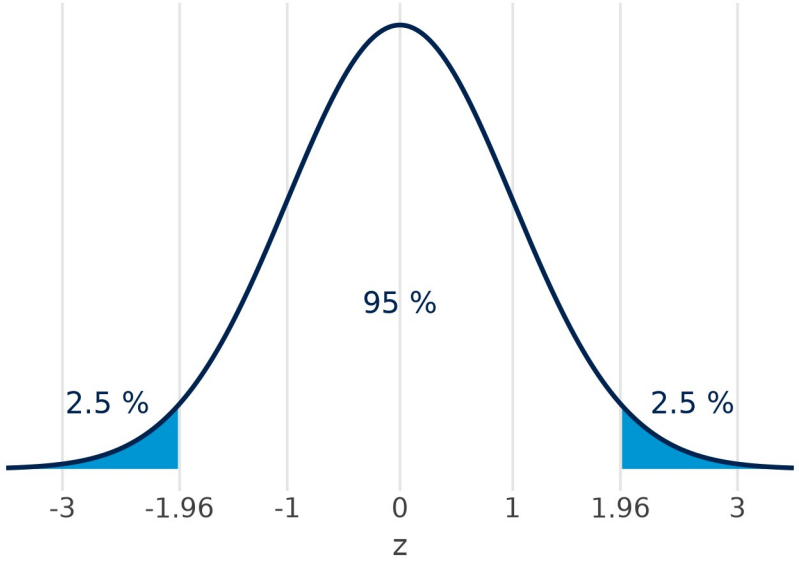
95 % ±2 sd, 60 to 100 bars: 42 of our 44 weeks

99.7 % ±3 sd, 50 to 110 bars: all 44 weeks

Our kiosk: 44 weeks at 3.00, mean 80, sd 10

Why it matters: averages and regression coefficients are approximately normal (central limit theorem), so we can test them.

The standard normal distribution



z	share below z
1.00	84.1 %
1.645	95.0 %
1.96	97.5 %
2.576	99.5 %

More than 90 bars? z = (90 − 80) / 10 = 1.0, so 100 − 84.1 = 15.9 % of weeks; our year: 6 of 44

1.96: the number behind 95 % confidence intervals and two-sided tests at 5 %.

Covariance and correlation

Part 4 of 11 · do two variables move together, and how strongly?

Covariance: do price and sales move together?



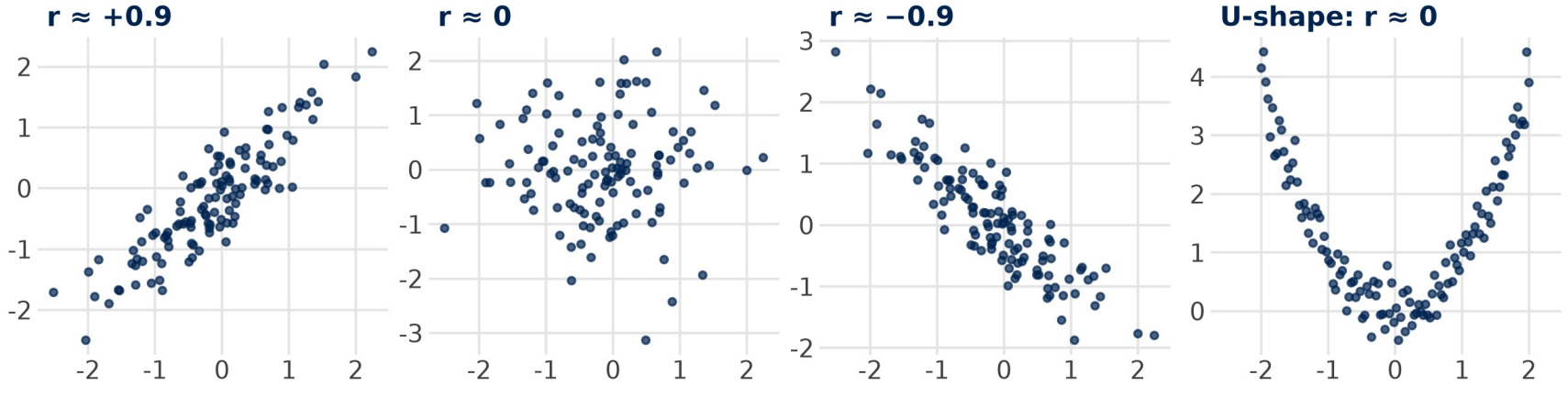
Week	p − p̄	x − x̄	product
1	−1.0	30	−30
2	−0.5	20	−10
3	0	0	0
4	0.5	−10	−5
5	1.0	−40	−40
Sum			**−85**

```
cov = −85 / (5 − 1) = −21.25
```

Sign = direction; the size depends on the units (EUR × bars).

Correlation: covariance without units

$$r = \text{cov} / (s_p \cdot s_x) = -21.25 / (0.79 \cdot 27.39) = -0.98 \qquad -1 \le r \le +1$$



Two warnings: r only measures straight-line relationships (the U has r ≈ 0), and correlation is not causation.

Sources: Bojinov, Parzen & Hamilton 2025 (HBS note 9-622-100)

From correlation to regression

Not causation: ice-cream sales and sunburn rise together; sunshine drives both

Correlation says how strongly two variables move together, from −1 to +1

Regression says by how much y changes when x changes by one unit

```
slope b = cov(p, x) / var(p) = −21.25 / 0.625 = −34 bars per EUR
line through the means: x = 80 − 34 · (p − 3)
```

Next: the least-squares line, built from exactly these pieces.

Python in Positron

Part 5 of 11 · mean, standard deviation, z-scores, covariance and correlation by hand, then checked; the kiosk's year

The packages we use today

Install (pip)	Load (import)	What for
polars	import polars as pl	data frames: read, select, calculate
plotnine	from plotnine import ggplot, aes, …	charts
great-tables	from great_tables import GT	formatted tables
statsmodels	import statsmodels.formula.api as smf	regression and tests
numpy	import numpy as np	np.log in model formulas
scipy	from scipy import stats	p-value of a correlation
pyarrow	no import needed	polars → plotnine and statsmodels
jupyter	no import needed	lets Quarto run Python



Install name ≠ import name: install great-tables, load great_tables. numpy, scipy and pandas come with statsmodels.

Install once, load in every file

1 Install once: in a Positron cell or the Console, with Python 3.14 (.venv)

```
%pip install polars plotnine great-tables statsmodels pyarrow jupyter
```

2 Load at the top of every file

```
import numpy as np
import polars as pl
import statsmodels.formula.api as smf
from great_tables import GT
from plotnine import (ggplot, aes,
                      geom_point, labs)
```

Install once per computer; the packages stay

Load every time Python starts or Quarto renders

Check: pl.__version__ prints a version number

'No module named …'? pick Python 3.14 (.venv) at the top right and install again; the terminal fix is in the set-up guide.

Step 1: the data and the mean

Python: polars

```
import polars as pl

kiosk = pl.DataFrame({
    "week":  [1, 2, 3, 4, 5],
    "price": [2.0, 2.5, 3.0, 3.5, 4.0],
    "units": [110, 100, 80, 70, 40],
})

n = kiosk.height                          # 5
mean_units = kiosk["units"].sum() / n     # by hand
kiosk["units"].mean()                     # check
kiosk["units"].median()
```

Result	Value
n	5
mean, by hand	80.0
mean(), check	80.0
median()	80.0

Run it: click into a cell and press Ctrl/Cmd + Enter; the result appears in the Console.

Step 2: variance, standard deviation, standard error

Python: by hand, then checked

```
dev = kiosk["units"] - mean_units
dev        # 30, 20, 0, -10, -40

var_units = (dev ** 2).sum() / (n - 1)   # 750.0
sd_units = var_units ** 0.5              # 27.39
kiosk["units"].std()                     # check: n - 1

se_units = sd_units / n ** 0.5           # 12.25
```

Result	Value
variance	750.0
sd, by hand	27.39
std(), check	27.39
standard error	12.25

∑ **Note:** std() divides by n − 1 (ddof=1), like the formula on the slide.

Step 3: standardise

Python: z-scores

```
kiosk = kiosk.with_columns(
    z_units=(pl.col("units") - pl.col("units").mean())
            / pl.col("units").std()
)
kiosk

kiosk["z_units"].mean()    # 0
kiosk["z_units"].std()     # 1
```

Week	Bars	z
1	110	1.10
2	100	0.73
3	80	0.00
4	70	−0.37
5	40	−1.46

Read it: week 5 lies 1.46 standard deviations below an average week.

Step 4: covariance and correlation

Python: by hand, then checked

```
dev_p = kiosk["price"] - kiosk["price"].mean()
dev_u = kiosk["units"] - kiosk["units"].mean()

cov = (dev_p * dev_u).sum() / (n - 1)          # -21.25
r = cov / (kiosk["price"].std()
           * kiosk["units"].std())             # -0.981

kiosk.select(
    pl.cov("price", "units").alias("cov"),
    pl.corr("price", "units").alias("r"))      # check
```

Result	Value
covariance	−21.25
correlation	−0.981
pl.cov, check	−21.25
pl.corr, check	−0.981



Same numbers as by hand: the computer only saves the typing.

Step 5: look at it

Python: plotnine

```
from plotnine import (ggplot, aes,
                      geom_point, labs)

(ggplot(kiosk, aes("price", "units"))
 + geom_point(size=3)
 + labs(x="Price (EUR)",
        y="Bars sold per week"))
```



Plot first: a correlation can hide a curve, an outlier or two groups.

Step 6: descriptive statistics and correlation with polars

Python: polars, then scipy for p

```
from scipy import stats

kiosk.select("price", "units").describe()

kiosk.select("price", "units").corr()   # matrix
kiosk.select(pl.corr("price", "units",
             method="spearman"))        # ranks
stats.pearsonr(kiosk["price"],
               kiosk["units"])          # r and p
```

Result	Value
mean, units	80.0
std, units	27.39
50 %, median	80.0
r, Pearson	−0.981
rho, Spearman	−1.000
p-value of r	0.003

Same numbers as by hand: describe() for every column at once, corr() for every pair; scipy adds the p-value.

The data: one year at the kiosk

Column	Meaning
week	1 to 52; weeks 1 to 5 are the price test of the slides
price	price per bar, EUR: 3.00 regular, 2.50 in four promotion weeks
units	bars sold in the week
temp	average temperature of the week, °C
display	yes if a display stood next to the till
rating	median customer rating of the week, 1 to 5 stars

File: `data/kiosk/kiosk_year.csv` 52 weeks, one row per week

Run the exercise with Posit Assistant

1. **Run first:** open refresher_exercise.qmd and run the first cell, so Posit Assistant sees the data frame kiosk
2. **Open the chat:** Posit Assistant icon in the activity bar, or Ctrl/Cmd + Shift + P → View: Show Posit Assistant
3. **Give context:** type @refresher_exercise.qmd, then one task per prompt: the file, the packages, the output
4. **Approve with care:** it asks before it runs code: Accept, Decline or reply; read every edit first
5. **Check:** compare its numbers with Steps 1 to 6; on an error, click Fix or Explain

Stay in Normal mode: never /yolo. Use /ask when you want an explanation without edits. You own every number.

Prompts for the exercise: one task at a time

Prompt for task 2

```
@refresher_exercise.qmd Task 2.
Use polars, not pandas. Mean and
median of units over all 52 weeks
and the median rating, with
kiosk.describe(). Write the code
into the Task 2 cell and explain
each line.
```

Prompt for task 5

```
Task 5. Correlation of units with
price and with temp in polars,
p-values with scipy.stats.pearsonr,
and a plotnine scatter of units
against temp. Run it, then explain
r and p for the kiosk owner.
```

Name three things: the file, the packages (polars, plotnine, scipy), the output

Try first: ten minutes on your own, then ask; compare both answers

Watch for slips: pandas instead of polars, np.std (divides by n), the wrong weeks

Your turn: one year at the kiosk, in Python

Pairs, in Python: 20 minutes → **Plenary: 5 minutes**

1. **Scales:** which scale is each of the six columns? What may you calculate for display and rating?
2. **Centre:** mean and median of units over all 52 weeks; the median rating
3. **Spread:** sd and standard error of units in the regular-price weeks (price 3.00)
4. **Normal:** how many regular weeks lie within mean ± 2 sd? z of the best regular week?
5. **Together:** r and p-value of units with price and with temp; plot units against temp

File: `sessions/01-foundations/refresher_exercise.qmd` (data, packages and the tasks)

Quiz 1: statistics in five questions

1 Satisfaction from 1 to 5 stars is measured on which scale?
A nominal B ordinal C ratio

2 Daily sales 10, 12, 11, 13, 54: what describes a typical day best?
A the mean (20) B the median (12) C the standard deviation

3 The kiosk records four times as many weeks. The standard error of the mean …
A halves B falls to a quarter C stays the same

4 A week's sales have z = −2. That week was …
A 2 bars below the mean B 2 standard deviations below the mean C 2 % below the mean

5 Price and bars sold have r = −0.98. What does this mean?
A higher prices cause lower sales B opposite directions, nearly linear C 0.98 bars fewer per EUR

3 minutes alone, then compare with your neighbour. Same format as the online quiz and the exam.

Refresher: key takeaways

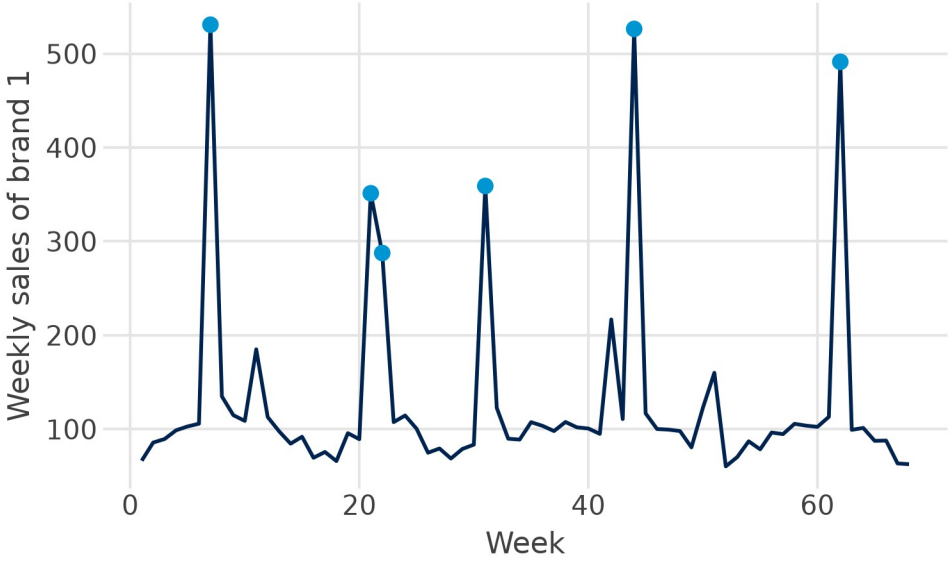
1. **Scale** first: it decides which statistics make sense
2. **Mean and median:** the median resists outliers; totals need the mean
3. **sd** describes the data; SE = sd / √n describes the precision of the mean
4. **$z = (x - \bar{x}) / s$** makes values comparable; 95 % of a normal lies within ±1.96 sd
5. **Covariance** gives the direction, correlation the strength (−1 to +1), regression the size

Next: linear regression on real chocolate scanner data, Parts 6 to 11.

Linear regression: theory

Part 6 of 11 · real data, simple and multiple regression, inference, fit, logs, from elasticity to price

The question: what does a price cut sell?



Weekly sales of brand 1; blue: weeks with a price cut (price below 1.20).

Data: 68 weeks of real scanner data, 4 brands

Normal week: price about 1.50, sales about 100

Price-cut week: price about 1.03, sales up to 530

Also in the data: competitor prices, features, displays, temperature, holidays

The manager asks: how much does 1 % lower price sell, and is the regular price right?

Correlation on real data: how strong is strong?

Rule of thumb for |r|

|r|	Interpretation
0 to 0.2	very weak
0.2 to 0.4	weak to moderate
0.4 to 0.6	medium to substantial
0.6 to 0.8	very strong
0.8 to 1.0	extremely strong

−0.94 sales and own price of brand 1: an extremely strong negative relationship

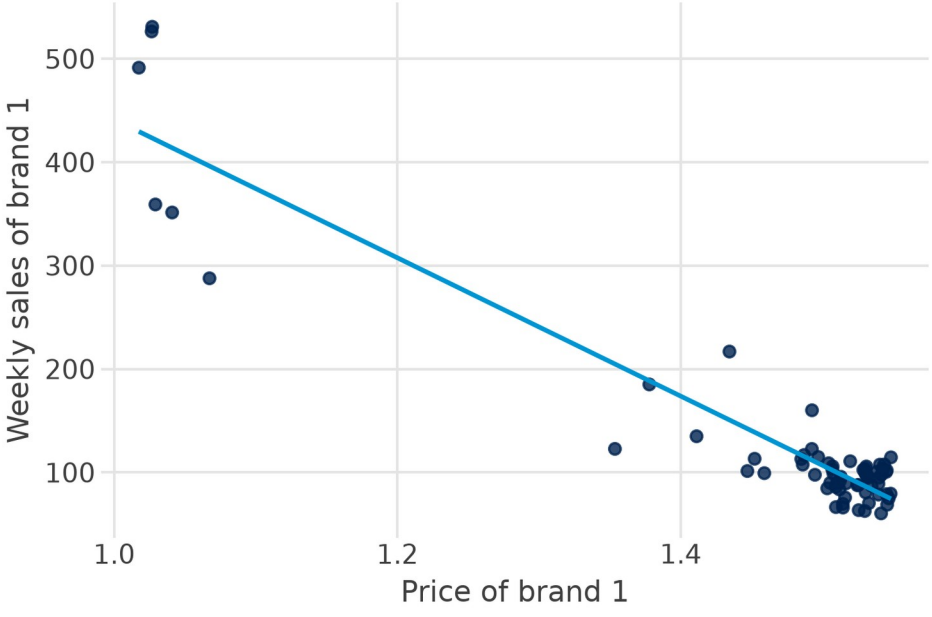
Kiosk: r = −0.98, five made-up weeks (Part 4)

Both: extremely strong, yet far from a perfect line

Real data are never a perfect line: competitors, features and the season move sales too.

Sources: Bojinov, Parzen & Hamilton 2025 (HBS note 9-622-100)

Simple linear regression




```
sales = b0 + b1 · price1 + e
```

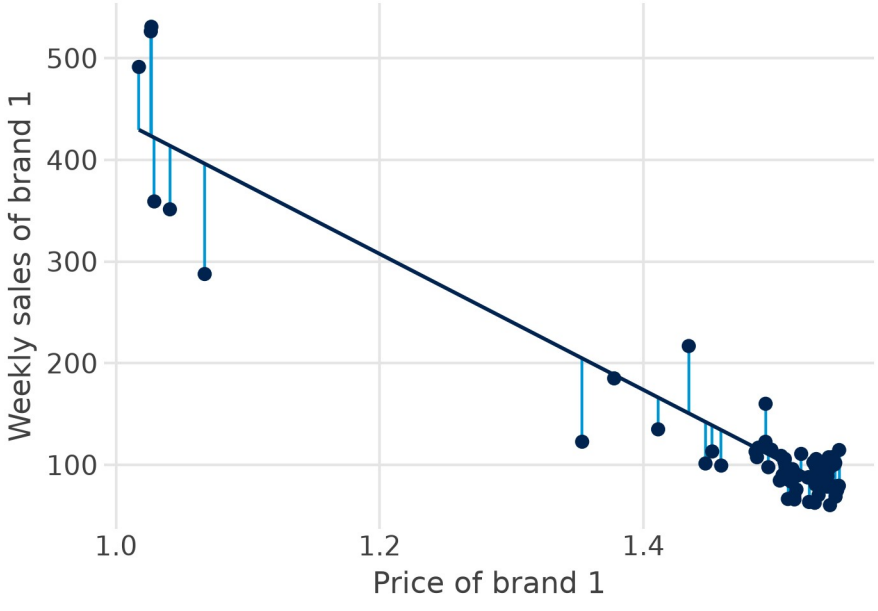
b0 = 1,110: intercept, sales at price 0 (not meaningful here)

b1 = −669: 10 cents more price, about 67 units fewer a week

e: residual, what the line does not explain

Look at the chart: the points curve, so a straight line misses the price-cut weeks.

Least squares: the line with the smallest squared errors




```
residual:  e_i = y_i − ŷ_i
OLS:       minimise Σ e_i²
```

Squared: misses above and below do not cancel

Large misses count much more than small ones

One line: the unique best fit for these data

Sources: Skiera, Reiner & Albers 2022; Bojinov, Parzen & Hamilton 2025

Multiple regression: holding other things constant

```
ln sales = b0 + b1·ln price1 + b2·ln price2 + b3·ln price3 + b4·ln price4
           + b5·feature1 + b6·temp + b7·december + e
```

Own mix

Price of brand 1

Feature ads, displays

Competitors

Prices of brands 2 to 4

Their features and displays

Context

Temperature

December and Easter weeks

Each b: the effect of one variable with all others held constant (ceteris paribus).

Left out a relevant variable? Its effect ends up in the others (omitted variable bias).

Is the effect real? t-test, p-value, interval, F-test

t-test

t = b / se

Own price: −3.64 / 0.30 = −12.1

p-value

Below 0.05: significant

Own price: $p < 0.001$; brand 4: $p = 0.86$

95 % confidence interval

About b ± 2 se

Own price: −4.24 to −3.04

F-test

Are all slopes together zero?

$F = 59.5$, $p < 0.001$

Significant is not important: read the size of b and its interval, not just the stars.

Sources: Bojinov, Parzen & Hamilton 2025; Skiera, Reiner & Albers 2022

How well does the model fit?

0.874 R^2: the model explains 87 % of the weekly variation in log sales

0.859 adjusted R^2: R^2 with a penalty for each extra variable

0.169 RMSE: the typical weekly miss, about 17 % in a log model

Careful

- R^2 never falls when you add a variable
- Compare R^2 only for the same outcome (sales vs log sales)
- High R^2 does not mean causal

Main model for brand 1: log-log, own and competitor prices, feature, temperature, December; 68 weeks.

Sources: Bojinov, Parzen & Hamilton 2025; Skiera, Reiner & Albers 2022

Functional forms and dummy variables

Model	Equation	b means	Brand 1
Linear	sales = a + b·price	+1 unit price: b units	b = −669 units
Log-log	ln sales = a + b·ln price	+1 % price: b %	b = −3.64 (elasticity)
Log-linear	ln sales = a + b·temp	+1 °C: about 100·b %	b = −0.0066: −0.66 %
Dummy in log model	ln sales = a + b·december	e^b − 1, vs other weeks	b = −0.385: −32 %

Why logs? sales respond in per cent, not in units, and the elasticity can be compared across brands and countries.

Sources: Skiera, Reiner & Albers 2022; Wooldridge 2025, ch. 6 to 7

From elasticity to a pricing decision

```
optimal price  p* = marginal cost × ε / (1 + ε)        (valid for ε < −1)
```

$52 Skiera et al.: ε = −2.34, unit cost $30 → p* = 30 × 2.34 / 1.34

×1.38 Brand 1: ε = −3.64 → price 38 % above marginal cost

Short term: promotion weeks inflate the elasticity; regular-price elasticities are smaller

Competitors react, and the retailer takes its margin

Sources: Skiera, Reiner & Albers 2022; Hanssens, Parsons & Schultz 2001

Assumptions and tests

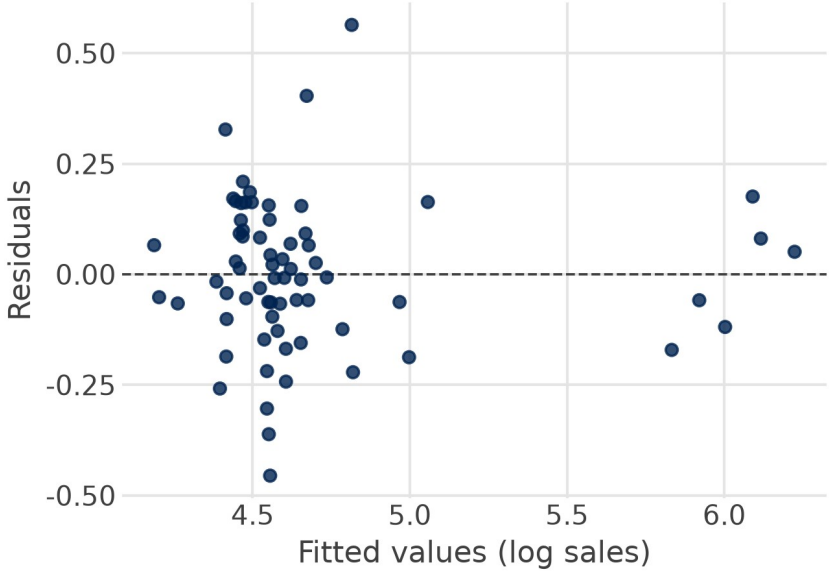
Part 7 of 11 · what OLS assumes, how to look, how to test, what to do

Seven assumptions behind OLS

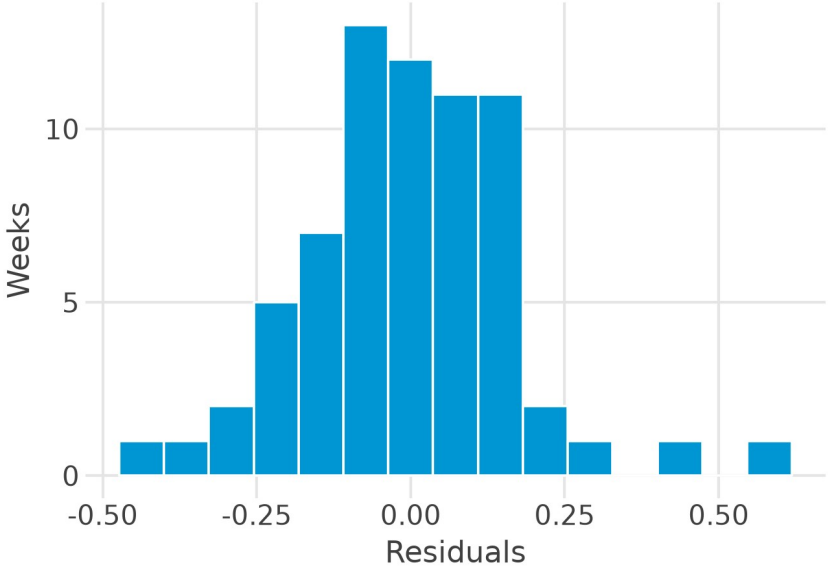
Assumption	If violated	Look or test (statsmodels)
1 Linear form	biased coefficients	residuals vs fitted; try logs
2 Exogeneity, no omitted variables	biased, not causal	theory, controls, experiments
3 Constant error variance	wrong standard errors	funnel in plot; Breusch-Pagan
4 No autocorrelation	standard errors too small	residuals over time; Durbin-Watson ≈ 2
5 Normal residuals	tests unreliable in small samples	histogram; Jarque-Bera
6 No multicollinearity	unstable b, large se	correlations; VIF below 10 (better 5)
7 Enough data, no dominant week	fragile results	n ≥ 3 to 5 × coefficients; Cook's distance

Sources: Skiera, Reiner & Albers 2022; Hair et al. 2019; Wooldridge 2025

Look first: two residual plots



Look for: a curve (wrong form) or a funnel (unequal variance)



Look for: a rough bell shape, no extreme weeks

Brand 1: no curve, no funnel, a roughly bell-shaped histogram.

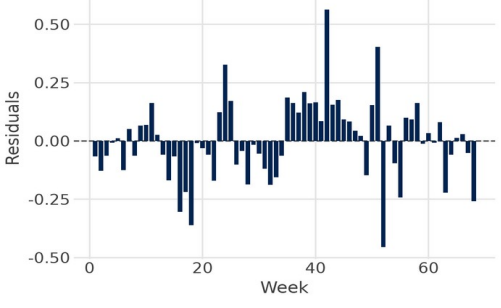
Then test: four numbers for the brand 1 model

1.41 Durbin-Watson: below 2, some autocorrelation between weeks

0.14 Breusch-Pagan p: no evidence of unequal variance

0.07 Jarque-Bera p: residuals roughly normal

2.4 largest VIF: no multicollinearity



Verdict: usable. Report Newey-West (HAC) standard errors because of the autocorrelation: 0.32 instead of 0.30 for own price.

When a test fails: what to do

Problem	Remedy	In statsmodels
Curve in residuals	logs, squares, interactions	np.log(x), I(x**2)
Unequal variance	robust standard errors	fit(cov_type="HC3")
Autocorrelation	HAC standard errors; lags; ARIMA errors (Session 4)	fit(cov_type="HAC", cov_kwds={"maxlags": 1})
Multicollinearity	more variation, combine variables, experiments	variance_inflation_factor
Dominant weeks	check data, report with and without	get_influence().cooks_distance
Endogeneity	experiments, instruments, fixed effects	Sessions 2 to 5

Sources: Skiera, Reiner & Albers 2022; statsmodels documentation

Three warnings from real data

Displays

- Only 4 display weeks
- VIF 11.4 and 10.9
- Effects cannot be separated

One Easter week

- A dummy for one week fits it exactly
- Leverage 1.00
- No real estimate of Easter

Mailings

- Skiera et al.: mailings and salespersons r = 0.989
- VIF 53
- Salesperson effect flips sign

The real fix: vary the marketing instruments independently, ideally in a field experiment.

Sources: Skiera, Reiner & Albers 2022; own calculation on the chocolate data

Endogeneity: when marketing follows demand

The assumption: explanatory variables are unrelated to the error term

Marketing breaks it: budgets go where managers expect success, so spend and unexplained sales rise together

Brand 1: price cuts come with features, so the price effect partly measures the feature

Evidence: in 15 Facebook experiments, regression often overstated the true lift, sometimes 3 times

Remedies: experiments (A/B, geo tests), instrumental variables such as cost shifters, panel data with fixed effects.

Sources: Skiera, Reiner & Albers 2022; Gordon, Zettelmeyer, Bhargava & Chapsky 2019

Regression in the real world

Part 8 of 11 · benchmarks, companies, what transfers to MMM

Benchmarks from decades of regressions

−2.62	mean price elasticity, 1,851 estimates (Bijmolt et al. 2005)
0.12	mean short-term advertising elasticity (Sethuraman et al. 2011)
−0.42	mean price elasticity in 14 Asia-Pacific countries, lower with high power distance (Datta et al. 2022)
−3.64	brand 1, chocolate: promotion-driven, many substitutes

Use benchmarks to judge whether your own estimate is plausible.

Where companies use regression

Meta Robyn

MMM in production: ridge regression on media

Booking.com

With Microsoft: CUPED, regression makes A/B tests faster

Google

Geo experiments: test vs control regions by regression

Walmart (M5)

Sales forecasts need price, promotion and event regressors

statworx

Frankfurt consultancy: price elasticities, log-log

Bayer

Budget across countries from estimated elasticities

Session 2: the marketing mix model is this regression with adstock and saturation.

Sources: Robyn docs; Deng et al. 2013; Vaver & Koehler 2011; Makridakis et al. 2022; Fischer et al. 2011

The chocolate data in Python

Part 9 of 11 · load and plot, the first regression, the log-log model, the checks

The data: 68 weeks of chocolate scanner data

Column	Meaning
week	1 to 68
sales	weekly unit sales of brand 1
price1 to price4	prices of brands 1 to 4
feature1 to feature4	share of stores with a feature ad, 0 to 1
display1 to display4	share of stores with an in-store display
fand1, fand3, fand4	share with feature and display together
temp	temperature, °C
december, easter	1 in December and Easter weeks

File: `data/legacy/chocolate_dataset.csv`

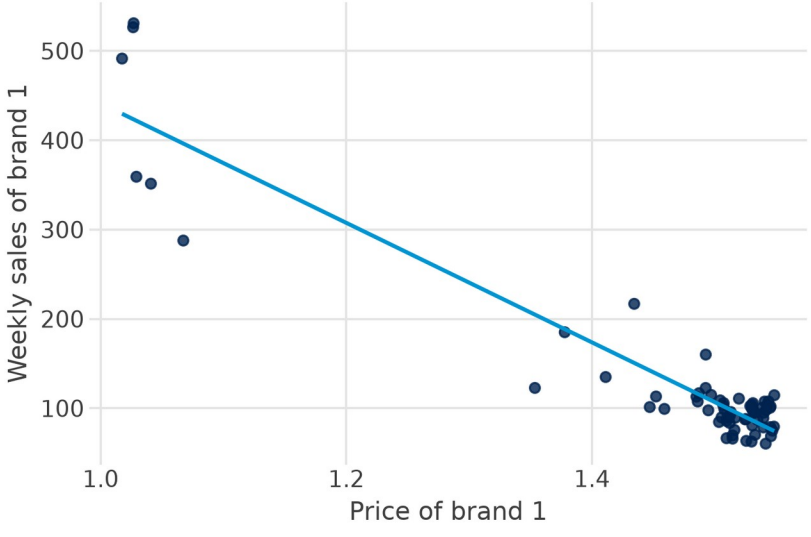
Step 1: load and look

Python: polars and plotnine

```
import polars as pl
from plotnine import (ggplot, aes, geom_point,
                      geom_smooth, labs)

choc = pl.read_csv(
    "data/legacy/chocolate_dataset.csv")
cols = ["sales", "price1", "price2",
        "price3", "price4"]
choc.select(cols).describe()
choc.select(cols).corr()

(ggplot(choc, aes("price1", "sales"))
 + geom_point()
 + geom_smooth(method="lm")
 + labs(x="Price of brand 1",
        y="Weekly sales"))
```



Look before you model: ranges, outliers, the shape of the relationship.

Step 2: the first regression

Python: statsmodels

```
import statsmodels.formula.api as smf

df = choc.to_pandas()
simple = smf.ols("sales ~ price1",
                 data=df).fit()
print(simple.summary())

simple.params      # b0, b1
simple.rsquared    # 0.884
```

	coef	std err	P>|t|
Intercept	1,110.0	44.0	0.000
price1	−668.8	29.8	0.000

R^2 = 0.884, n = 68

Read it: 10 cents more price, about 67 units fewer per week; the price explains 88 % of the weekly variation.

But: the residuals curve. Next: the log-log model with the marketing mix.

Step 3: the log-log model with the marketing mix

Python: statsmodels

```
import numpy as np

model = smf.ols(
    "np.log(sales) ~ np.log(price1)"
    " + np.log(price2) + np.log(price3)"
    " + np.log(price4) + feature1"
    " + temp + december",
    data=df).fit()
print(model.summary())
```

	coef	p
ln price1	−3.64	0.000
ln price2	−0.80	0.076
ln price3	0.36	0.109
ln price4	−0.03	0.855
feature1	0.18	0.307
temp	−0.007	0.045
december	−0.38	0.009

R^2 = 0.874, adjusted 0.859, n = 68

Read it: 1 % higher price, 3.6 % fewer sales; December weeks 32 % lower; each degree warmer 0.7 % lower.

Step 4: check the assumptions

Python: tests and robust errors

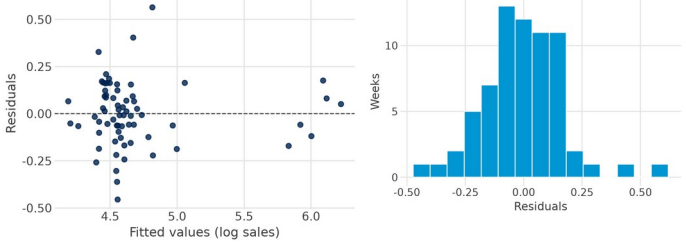
```
from statsmodels.stats.stattools import (
    durbin_watson, jarque_bera)
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor)

durbin_watson(model.resid)                      # 1.41
het_breuschpagan(model.resid, model.model.exog)[1]  # 0.14
jarque_bera(model.resid)[1]                     # 0.07
X = model.model.exog
[variance_inflation_factor(X, i) for i in range(1, 8)]

hac = model.model.fit(cov_type="HAC",
                      cov_kwds={"maxlags": 1})
```

Residual plot

```
from plotnine import geom_hline
df["fit"] = model.fittedvalues
df["res"] = model.resid
(ggplot(df, aes("fit", "res"))
 + geom_point()
 + geom_hline(yintercept=0))
```



Decide: report the HAC result; the own-price elasticity stays −3.64, its standard error rises to 0.32.

Quiz 2: regression in five questions

1 What does OLS minimise?
A the sum of residuals B the sum of absolute residuals C the sum of squared residuals

2 In ln sales = a + b · ln price, b = −3.64 means …
A 1 EUR more, 3.64 units fewer B 1 % higher price, 3.6 % fewer sales C 3.64 fewer per week

3 A VIF of 11 signals …
A autocorrelation B multicollinearity C heteroscedasticity

4 Weekly data, Durbin-Watson = 1.41. This suggests …
A positive autocorrelation B no problem C unequal variance

5 R^2 rises from 0.85 to 0.86 after adding a variable. This shows …
A the variable matters B nothing: R^2 never falls C the model is causal

3 minutes alone, then compare with your neighbour. Same format as the online quiz and the exam.

In-class assignment

Part 10 of 11 · groups of 3, 25 minutes, on the chocolate data

In-class assignment: pricing brand 1

Groups of 3: 20 minutes → **Plenary: 5 minutes**

1. **Competition:** plot sales against price2, price3 and price4. Which brand competes most?
2. **Displays:** add display1 and fand1 to the log-log model. What happens to the VIFs and coefficients, and why?
3. **Price:** marginal cost is 0.80. What price does $\varepsilon = -3.64$ suggest? Why is the real price about 1.50?
4. **The manager:** write one sentence for the brand manager, with a number and one caveat.

File: `sessions/01-foundations/regression_chocolate.qmd` (the code of Part 9 and these questions)

Coding exercise 1

Part 11 of 11 · individual, 5 % of the grade

Coding exercise 1: what to do

1. **Explore:** a great_tables summary and two plotnine charts: sales over time, sales against price1
2. **Simple model:** sales ~ price1; interpret the slope for a 10-cent price change
3. **Log-log model:** own and competitor prices, feature1, display1, temp, december; interpret 3 coefficients
4. **Assumptions:** residual plot, histogram, Durbin-Watson, Breusch-Pagan, Jarque-Bera, VIF; one sentence each
5. **Robust errors:** refit with HAC standard errors; what changes and what does not?
6. **Recommend:** the optimal price for marginal cost 0.90, with two caveats, in at most 100 words

Brief: `assignments/coding-exercises/exercise-1.qmd` and on Canvas

Coding exercise 1: submission and grading

Submit on Canvas

- exercise1_lastname.qmd and the rendered .html
- Renders from top to bottom without errors
- Every number in the text from code
- Prompt log at the end: what you asked, what you changed
- Deadline: see Canvas

Criterion	Points
Code runs, charts and tables clean	5
Interpretation of the coefficients	6
Assumption checks and conclusion	5
Recommendation with caveats	2
Prompt log	2
Total (5 % of the grade)	20

Individual work: discuss ideas, but write and check your own code. AI assistants are allowed; you own every number.

Regression: key takeaways

1. **OLS** draws the line with the smallest squared errors; each b holds the other variables constant
2. **Logs** turn coefficients into elasticities: brand 1, −3.64
3. **Look, then test:** residual plots, Durbin-Watson, Breusch-Pagan, Jarque-Bera, VIF
4. **Fix what fails:** robust or HAC errors, better form, more variation; experiments for causality
5. **Decide:** $p^* = \text{cost} \times \varepsilon / (1 + \varepsilon)$, with the caveats

Next session: the marketing mix model, a regression with adstock and saturation.

References: statistics refresher

Books and readings

Field, A. (2024). Discovering Statistics Using IBM SPSS Statistics, 6th ed. SAGE. Chapters 1 to 2 and 8.

Hair, J. F. et al. (2019). Multivariate Data Analysis, 8th ed. Cengage. Chapter 2.

Stock, J. H. & Watson, M. W. (2019). Introduction to Econometrics, 4th ed. Pearson. Chapters 2 to 3.

Online

Seeing Theory. Brown University: interactive visual introduction to probability and statistics (free).

StatQuest. Videos on the normal distribution, standard deviation vs standard error, covariance and correlation (free).

polars documentation. mean, median, std, cov, corr.

Gelman, Hill & Vehtari (2020). Regression and Other Stories, ch. 3 to 4 (free PDF).

References: linear regression

Academic

Bojinov, I. I., Parzen, M. & Hamilton, P. J. (2025). Linear Regression. HBS note 9-622-100 (Canvas).

Skiera, B., Reiner, J. & Albers, S. (2022). Regression analysis. In Homburg et al. (eds), Handbook of Market Research, 299–327. Springer.

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Hanssens, D. M., Parsons, L. J. & Schultz, R. L. (2001). Market Response Models. Kluwer.

Bijmolt, T., van Heerde, H. & Pieters, R. (2005). Journal of Marketing Research 42(2).

Gordon, B. R. et al. (2019). Marketing Science 38(2).

Learning resources

James, G. et al. (2023). ISLP, ch. 3 and the Python lab (free).

Schwarz, Chapman & Feit (2020). Python for Marketing Research and Analytics, ch. 7. Springer.

statsmodels. Documentation and example notebooks.

Turrell, A. Coding for Economists: regression and diagnostics (free).

StatQuest. Linear regression, clearly explained (video).

DataCamp. Introduction to Regression with statsmodels in Python.

VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

INSTITUTE FOR INTERNATIONAL BUSINESS
Welthandelsplatz 1, 1020 Vienna, Austria

DR ARNE FLOH
Senior Lecturer in Marketing

F +43-31336-6367
arne.floh@wu.ac.at
www.wu.ac.at/en/welthandel/arne-floh
Office hours: book online